



GRAPH-BASED REAL-TIME UPI FRAUD DETECTION AND SCAM LINK / QR IDENTIFIER SYSTEM USING AI AND ML

Batch No: < >

<Register No.>

<Register No.>

<Register No.>

<Register No.>

Guide Details:

<Project Supervisor>

<Designation>



OUTLINE

Abstract

Introduction

Literature Survey

Research Gaps

Objectives

Methodology

Architecture (Explain the modules in detail)

Results

Conclusion & Future Scope

References



ABSTRACT

UPI settled 24.51 billion payments worth Rs 29.82 trillion in August 2026. A UPI payment is irrevocable, so a system that classifies stored transactions describes a loss that has already happened; and in the dominant Indian fraud pattern the account holder authorises the payment themselves, so nothing about the payer's behaviour is anomalous.

This project assesses a payee BEFORE money moves. A scanned QR code, a pasted UPI ID or a phone number is scored by ten independent evidence families - address integrity, payee reputation, payer behaviour, relationship, amount, stated purpose, message pressure, link safety, a calibrated classifier and the payee's position in an incrementally maintained payment graph.

Missing evidence is never treated as favourable evidence. Every assessment publishes a risk score, a confidence score from evidence coverage, and one verdict: APPROVE, WARN, STEP_UP or BLOCK.

On a 120,000-row simulator the calibrated model reaches PR-AUC 0.486 and 49.3% recall at a 0.95% false-positive rate. The data is simulated and the system holds no NPCI, PSP or banking integration.



INTRODUCTION

- UPI carries 24.51 billion payments a month - 791 million a day - and settles them in seconds.
- A UPI transfer is irrevocable. There is no chargeback: once the payer approves, recovery depends on freezing an account that is usually already empty.
- Most reported UPI fraud in India is authorised by the victim: a fake customer-care call, a refund that needs a fee first, a QR code presented as a way to receive money.
- Detection after settlement therefore describes losses instead of preventing them.
- Modelling only the payer cannot see an authorised payment - the payer is behaving normally by definition, at an ordinary hour, on their own device.
- What is left to examine is the counterparty: who is being paid, what is known about that address, and what arrived alongside the request.



INTRODUCTION

What the system does

- Scores a payee address before any money moves, within a few hundred milliseconds, using ten independent evidence families.
- Distinguishes four states of knowledge - UNKNOWN, OBSERVED, REGISTERED and VERIFIED - and never claims a verification it did not perform.
- Reports two numbers, not one: how bad the available evidence looks, and how much evidence there was.
- Reads a QR from the camera or an uploaded image, checks scam links, look-alike bank handles and Unicode confusables, and scores coercive framing in the message.
- Delivered as a React web application, an Express and FastAPI service, one SQLite store and a Capacitor Android build - 581 automated tests.



LITERATURE SURVEY

Research Paper	Method / Focus	Relevance
Lopez-Rojas, Elmir & Axelsson, "PaySim: A Financial Mobile Money Simulator for Fraud Detection", EMSS, 2016	Agent-based simulation of mobile-money transfers; fraud injected as agent behaviour rather than as a rule over features.	Our simulator follows the same principle - labels come from four fraud behaviours - and extends it by also modelling what the deployment cannot observe.
Weber et al., "Anti-Money Laundering in Bitcoin: Graph Convolutional Networks for Financial Forensics", KDD Workshop, 2019	Elliptic transaction graph; logistic regression, random forest and GCN compared on illicit-node classification.	Confirms graph structure adds signal over local features, and that gradient boosting stays a strong baseline - which is what we use.
Hooi et al., "FRAUDAR: Bounding Graph Fraud in the Face of Camouflage", ACM SIGKDD, 2016	Dense-subgraph detection on a bipartite payer-payee graph with provable bounds under camouflage.	Whole-graph and batch. Motivates our incremental union-find and per-payee cache, which answer in a lookup on the live path.



LITERATURE SURVEY

Research Paper	Method / Focus	Relevance
Dou et al., “Enhancing GNN-based Fraud Detectors against Camouflaged Fraudsters (CARE-GNN)”, ACM CIKM, 2020	Label-aware neighbour sampling and reinforcement-learned neighbour selection on relation graphs.	Assumes node features are always present; our design treats an unobservable payee as a first-class state instead.
Liu, Ting & Zhou, “Isolation Forest”, IEEE ICDM, 2008	Unsupervised anomaly detection by random isolation; linear time, low memory.	Used as our unsupervised second opinion, fitted on legitimate traffic only (ROC-AUC 0.854).
Ke et al., “LightGBM: A Highly Efficient Gradient Boosting Decision Tree”, NeurIPS, 2017	Histogram-based gradient boosting; native missing-value handling by learned default split directions. Department of Information Technology	We use histogram gradient boosting and reuse native NaN handling for a different purpose: representing evidence that could not be measured.



LITERATURE SURVEY

Research Paper	Method / Focus	Relevance
Niculescu-Mizil & Caruana, "Predicting Good Probabilities with Supervised Learning", ICML, 2005	Platt scaling versus isotonic regression across classifier families; reliability diagrams.	Boosted trees are over-confident, so our classifier is wrapped in isotonic calibration before its score is published.
Davis & Goadrich, "The Relationship Between Precision-Recall and ROC Curves", ICML, 2006	Formal relation between the two curve families under class skew.	Why we report PR-AUC and recall at a fixed false-positive budget rather than accuracy at a 1% base rate.
Saito & Rehmsmeier, PLOS ONE, vol. 10, no. 3, 2015	Empirical comparison of evaluation metrics under extreme class imbalance.	Supports our operating point: recall measured inside a 1% false-positive budget, chosen on validation.



LITERATURE SURVEY

Research Paper	Method / Focus	Relevance
Lundberg & Lee, “A Unified Approach to Interpreting Model Predictions (SHAP)”, NeurIPS, 2017	Additive feature attribution with Shapley values against a background distribution.	Used for per-payment explanation; the background is sampled from training traffic, not from noise.
Toh, “Combating Authorized Push Payment Scams in Fast Payment Systems”, Federal Reserve Bank of Kansas City, 2024	Policy review of APP scams across fast-payment rails: confirmation of payee, payment friction.	Establishes that counterparty signals and pre-payment friction are the effective levers - the premise of this project.
“Detection of Mule Accounts and Fraudsters in UPI Transactions using AI and ML”, IRE Journals, 2024	Supervised classification over transaction attributes to flag mule accounts in UPI-style logs. Department of Information Technology	Post-hoc and ledger-wide. We read the same inflow shape from contributed data, before the payment, and say so when we cannot.



RESEARCH GAPS

1. Detection happens after settlement

Almost all reviewed work scores a stored transaction record; on an irrevocable rail that describes a completed loss.

2. Payer-only modelling cannot see authorised fraud

In a social-engineering payment there is no deviation from the payer's norm to describe.

3. Missing information is replaced by favourable defaults

Imputing an unknown payee as “moderately established” makes a mule account - which has no history by definition - look safe.

4. One number conflates risk with certainty

A single percentage cannot separate a well-supported all-clear from an assessment that could not be made.

5. Correlated evidence is counted as independent agreement

Two modules applying the same night-hour test to the same timestamp are one observation, not two.

6. Graph analytics are batch, and claims exceed available data

Dense-block and GNN methods need the whole graph, and payer-side proposals often assume rail-level access.



OBJECTIVES

1. Assess a payee address before any money moves, from a QR code, a UPI ID or a phone number.
2. Model knowledge explicitly: UNKNOWN, OBSERVED, REGISTERED and VERIFIED, with verification only from a provider.
3. Score ten independent evidence families and merge them with a damped-maximum rule plus fact-aware corroboration.
4. Preserve availability end to end - NaN for unmeasurable, zero for measured-as-none, null on the wire.
5. Publish risk and confidence separately, and one verdict vocabulary from one mapping.
6. Train a calibrated classifier on behaviour-derived labels; report PR-AUC, recall at a 1% FPR budget and per-scenario recall.
7. Maintain graph intelligence incrementally so the real-time path never rebuilds a graph.
8. Deliver as a web application and an Android build, with an assistant that explains but cannot decide.



METHODOLOGY

1. Entity status - what the system is allowed to claim

UNKNOWN: never observed, and reported as such. OBSERVED: seen in contributed data.

REGISTERED: an account exists - which is not safety. VERIFIED: a provider confirmed it exists.

A payee with fewer than five distinct payers has no readable shape: a repeat ratio over two payers is arithmetic, not measurement.

2. Ten independent evidence families

payee_history (24), relationship (18), payer_behaviour (16), ml_classifier (12), address_and_qr (11), graph (8), amount_context (5), message_pressure (3), link_safety (2), stated_intent (1).

Each family returns a score, an availability flag, a severity, a message and the typed facts its score rests on. The weights are confidence weights: how much less would we know without this family.



METHODOLOGY

1. Availability: zero and unknown are different inputs

0.0 means we looked and the answer is none; NaN means we could not look. A companion feature states which case holds, and an unavailable family is refused a score at construction time.

2. Combination - the strongest signal leads

risk = ordered[0] + 0.4 x sum(rest), capped at 99. Addition saturates on any two moderate signals; a plain maximum throws corroboration away.

3. Fact-aware corroboration

bonus = AGREEMENT_BONUS[min(distinct facts, reporting families)], with {2:6, 3:14, 4:22, 5:28, 6:32}. Counting modules paid a bonus for reading one clock twice.

4. One verdict mapping

BLOCK at 70, STEP_UP at 45, WARN at 22, otherwise APPROVE - in one module. Hard-block findings override the arithmetic; the legacy LOW/MEDIUM/HIGH label is derived from the band so the two cannot disagree.



METHODOLOGY

1. Dataset - labels from behaviour, not from a rule

120,000 payments, 2,000 synthetic payers, 1.2% fraud across four scenarios: social engineering 50%, mule collection 22%, account takeover 20%, card testing 8%. Classes overlap deliberately - legitimate payments reach the Rs 1 lakh cap.

2. Features and the observation mask

Twelve features; payee history is hidden on 24.5% of rows with a probability that rises with payer count, so unavailability is a case the model has learned rather than a gap papered over.

3. Model, calibration and operating point

HistGradientBoosting (300 iterations, balanced) with isotonic calibration; 72k / 18k / 30k stratified split; the threshold that maximises recall inside a 1% false-positive budget is chosen on validation and measured once on test.

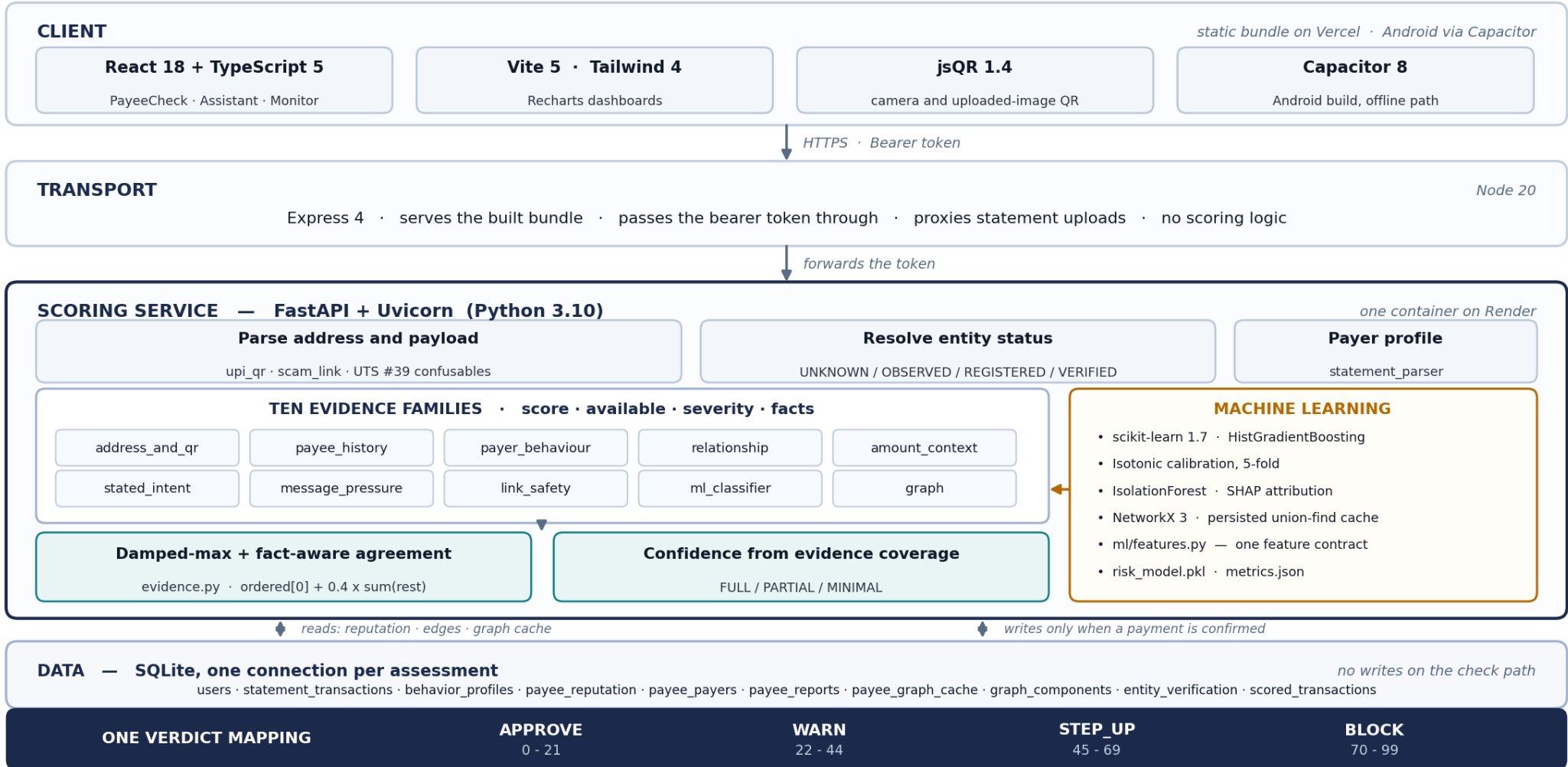
4. Incremental graph intelligence

A persisted union-find table and a per-payee cache are updated when a payment is confirmed; the read path answers a bounded two-hop overlap query with no writes and no rebuild.



ARCHITECTURE

System Architecture — pre-payment risk assessment, completed before any money moves





RESULTS

Metric	Logistic baseline	Selected model	Reading
ROC-AUC	0.9410	0.9652	Optimistic at a 1% base rate
PR-AUC	0.3037	0.4860	The informative view under imbalance
Recall at 1% FPR budget	27.89%	49.30%	Fraud caught inside the budget
Realised FPR	0.97%	0.95%	Measured on the test split
Precision at that point	25.52%	38.38%	About two alerts per true positive
Brier score	0.00942	0.00798	Base-rate reference is 0.01169
Brier skill score	0.1944	0.3174	Calibration gain over that reference



RESULTS

Fraud scenario	n	Baseline	Our model	Character of the case
account_takeover	73	78.08%	94.52%	Burst to fresh payees, amount pushed up
card_testing	20	100%	100%	Tiny probe payments in quick succession
mule_collection	80	13.75%	63.75%	Young payee, many one-time payers
social_engineering	182	6.04%	19.23%	The payer authorises it - nothing about the payer is wrong
Payer features only	-	PR-AUC 0.3175	SE recall 3.85%	Ablation: one half of the feature set
Payee features only	-	PR-AUC 0.1495	SE recall 11.54%	Ablation: the other half
Both halves together	-	PR-AUC 0.4860	SE recall 19.23%	+7.7 points over the better half - the argument for the payee check



RESULTS

Case	Payment	Amount (Rs)	Model p	Risk	Verdict	What decided it
--	Canteen lunch / groceries / cab	90 - 640	0.001	0 - 12	APPROVE	Ordinary payments are cleared
--	Monthly rent	18,500	0.000	20	APPROVE	83x this payer's median, still approved
F1	Probe burst, 3rd of 8	5	0.024	14	APPROVE	Open item: the 3rd probe of a 21-minute run is not yet the burst the model was trained on
F2	Funnel account, 3rd of 5	4,500	0.325	99	BLOCK	Carried by the model
F3	Hijacked profile, 2nd of 3	92,400	0.007	74	BLOCK	Model below threshold - payee evidence decided
F4	Victim-authorized transfer	18,000	0.007	58	STEP_UP	Model below threshold - payee evidence decided
F5	Look-alike bank handle	47,500	0.220	99	BLOCK	Carried by the model



CONCLUSION & FUTURE SCOPE

Conclusion

- A pre-payment assessment is possible on the information a payer-side system can actually gather, and it stops payments a payer-behaviour model cannot see: F3 and F4 are stopped while the classifier scores them at 0.007.
- Absence of evidence is never evidence of safety - risk and confidence are separate numbers, and an unavailable family carries no score anywhere in the system.
- Measured, not asserted: PR-AUC 0.486, 49.3% recall at 0.95% FPR, 19.2% on victim-authorized fraud, and a Rs 5 probe payment that is still missed, reported rather than tuned away.

Future scope

- Replace the simulator with real labelled data - the twelve-column feature contract trains the pipeline unchanged.
- Model the payee's inflow as a sequence; evaluate a graph neural network once the contributed graph is large enough.
- Confirmation of payee through a provider, privacy-preserving reputation sharing between deployments, and on-device scoring for the Android build.



REFERENCES

1. NPCI, UPI product statistics, 2026; UPI August 2026: 24.51 bn transactions, Rs 29.82 trn.
2. NPCI, Revision in UPI transaction limits for person-to-merchant categories, effective 15 September 2025.
3. Y. Z. Toh, "Combating authorized push payment scams in fast payment systems," Federal Reserve Bank of Kansas City, 2024.
4. E. A. Lopez-Rojas, A. Elmir and S. Axelsson, "PaySim: a financial mobile money simulator for fraud detection," EMSS, 2016.
5. M. Weber et al., "Anti-money laundering in bitcoin: graph convolutional networks for financial forensics," KDD Workshop, 2019.
6. B. Hooi et al., "FRAUDAR: bounding graph fraud in the face of camouflage," ACM SIGKDD, 2016.
7. G. Ke et al., "LightGBM: a highly efficient gradient boosting decision tree," NeurIPS, 2017.
8. A. Niculescu-Mizil and R. Caruana, "Predicting good probabilities with supervised learning," ICML, 2005.
9. T. Saito and M. Rehmsmeier, PLOS ONE, vol. 10, no. 3, 2015; J. Davis and M. Goadrich, ICML, 2006.
10. S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," NeurIPS, 2017.
11. F. T. Liu, K. M. Ting and Z.-H. Zhou, "Isolation forest," IEEE ICDM, 2008.
12. Unicode Consortium, "Unicode security mechanisms," Unicode Technical Standard #39.